

# Construction of Shanks-type $p$ -Series Convergence Accelerators

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## Introduction

The *Shanks Transformation* is a delightful numerical method for *convergence acceleration*: it's a simple transformation on a sequence  $x_n \rightarrow x$  that attempts to more accurately predict the limiting value  $x$  (what we desire) by estimating the error of the sequence data  $x_n$  at each step  $n = 1, 2, \dots$ .

More precisely, we know that at each step  $x_n = x + \epsilon_n$  can be written in terms of some error  $\epsilon_n \rightarrow 0$  and we'll assume an *error model* of the form  $\epsilon_n = C\alpha^n$ : which is to say we'll assume that  $x_n$  converges to  $x$  at a roughly geometric rate. The big idea is that by using the values of  $x_n = x + C\alpha^n$ ,  $x_{n+1} = x + C\alpha^{n+1}$ , and  $x_{n+2} = x + C\alpha^{n+2}$  we should be able to *solve* for the unknown parameters  $x, C, \alpha$  and simply output the value of  $x$  we predict:

$$x_{n+2} - \frac{x_{n+2} - x_{n+1}}{1 - 1/\alpha} = x + C\alpha^{n+2} - \frac{C\alpha^{n+2} - C\alpha^{n+1}}{1 - 1/\alpha} = x$$

Where to find  $\alpha$  we can take a ratio  $\frac{x_{n+2} - x_{n+1}}{x_{n+1} - x_n} = \frac{C\alpha^{n+2} - C\alpha^{n+1}}{C\alpha^{n+1} - C\alpha^n} = \alpha$ , and putting the two together we have;

$$S_n = x_{n+2} - \frac{x_{n+2} - x_{n+1}}{1 - \left(\frac{x_{n+2} - x_{n+1}}{x_{n+1} - x_n}\right)^{-1}} = \frac{x_{n+2}x_n - x_{n+1}^2}{x_{n+2} - 2x_{n+1} - x_n} \approx x$$

And once we have the nice formula, we can forget about  $C, \alpha$  and use  $S_n$  directly to predict the value of  $x$ : even if  $x_n$  is not *exactly*  $x + C\alpha^n$  (which it rarely will be), the formula still very accurately approximates geometric convergence.

I'll let you explore how effective this simple formula is hands on!

*Exercise:* The “Archimedean formula” for estimating  $\pi$  is described by the sequence  $x_n = 3 \times 2^n \sin(\frac{\pi}{3 \times 2^n})$  for  $n = 1, 2, \dots$ : the result of bisecting an inscribed hexagon into finer and finer polygons and taking their circumference.

Prove that  $\lim_{n \rightarrow \infty} x_n = \pi$ . Then, using  $\sin(x) \sim x - \frac{x^3}{6} + \dots$ , demonstrate that the error  $\epsilon_n$  indeed has the desired form of  $C\alpha^n$ .

| $n$ | $x_n$       | $S_n$       | $S_n^2$ |
|-----|-------------|-------------|---------|
| 1   | 3           | 3.141717033 | ?       |
| 2   | 3.105828541 | ?           | ?       |
| 3   | 3.132628613 | ?           | ?       |
| 4   | 3.139350203 | ?           | ?       |
| 5   | 3.141031951 | ?           |         |
| 6   | 3.141452472 | ?           |         |
| 7   | 3.141557608 | ?           |         |
| 8   | 3.141583892 |             |         |
| 9   | 3.141590463 |             |         |

Now, open up **Excel** or **GoogleSheets** and make a column for  $x_n$ . Using three consecutive cells of  $x_n$ , make a column for  $S_n$ . Finally, drag that  $S_n$  column one place to the right to generate a column for  $S_n^2$ : the application of the Shanks transform to itself.

What do you observe about the accuracy of the  $\pi$ -estimates? Make a quantitative comparison.

What I'm interested in is what happens when you alter the error model we used at the start: in particular what the resulting series transformation should be when  $\epsilon_n = \frac{C}{(n+k)^p}$  and where  $C, k, p$  are unknown. This kind of “ $p$ -series error” comes up very frequently in practice:

- *p-Series:* Sums which look like  $\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}$  or  $\frac{\pi}{4} = \sum_{n \geq 0} \frac{(-1)^n}{2n+1}$  or  $\ln(2) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n}$  or things of this kind are limits of the partial sums  $x_n = \sum_{k \leq n} a_k$  whose error corresponds to the tail  $\epsilon_n = \sum_{k > n} a_k$  which decays like  $1/\text{some power of } n$ .
  - See the Euler–Maclaurin summation formula Wikipedia page for several examples of exactly why these series converge with the error claimed.
- *Numerical Integration:* Approximating an integral  $\int_a^b f(t)dt$  by a weighted sum of  $n$  evenly spaced evaluation points famously converges rather slowly: at a rate of  $O(1/n)$ . More advanced methods of (Newton) quadrature such as the Trapezoidal rule or Simpson's rule significantly improve the exponent  $p$  (respectively 3 and 5) but do not change the fact that the error is a  $p$ -series.
- *Monte Carlo Estimation:* If you want to estimate the mean  $\mu$  of a random variable  $X$  and you take  $n$  samples  $X_1, \dots, X_n$ , you would predict  $\mu \approx \frac{X_1 + \dots + X_n}{n}$ . The *variance* of that guess would be  $\frac{\text{Var}(X) + \dots + \text{Var}(X)}{n^2} = \frac{1}{n} \sigma^2$  and hence we typically say that this estimator has an error (standard deviation) of  $\epsilon_n \sim \frac{\sigma}{\sqrt{n}}$ . So,  $p = 1/2$  here.
- *Miscellaneous Limits:* Things like  $x_n = \sum_{k=1}^n \frac{1}{k} - \ln n \rightarrow \gamma$  or  $x_n = n \sin(\pi/n) \rightarrow \pi$  or  $x_n = \sqrt{n+a} - \sqrt{n+b} \rightarrow 0$  converge according to a  $p$ -series error in  $n$ . In general, unless there is an iterative process or an exponential in sight,  $O(1/n^p)$  is probably going to be the error.

*Exercise:* Convince yourself that a typical rectangular approximation to an integral should have error  $O(1/n)$  by writing  $\int_0^1 f(t)dt = \sum_{k=0}^{n-1} \int_{k/n}^{(k+1)/n} f(t)dt$ . Think about the mean value theorem to compare  $\int_{k/n}^{(k+1)/n} f(t)dt$  with  $\frac{1}{n} f(t_k^*)$  for some  $t_k^* \in [k/n, (k+1)/n]$ . What is the constant  $C$  hiding in  $O(1/n)$ ? Feel free to compute some sums numerically!

## Formula Development

To get started, we'll assume  $x_n = x + \frac{C}{(n+k)^p}$ , and we'll make use of the same ratio-of-differences technique used at the start;

$$\delta_n = \frac{x_{n+1} - x_n}{x_{n+2} - x_{n+1}} - 1 = \frac{C/(n+1+k)^p - C/(n+k)^p}{C/(n+2+k)^p - C/(n+1+k)^p} - 1 \sim \frac{p+1}{n+k}$$

Which can be seen from the fact that  $\frac{1}{(n+\square)^p} - \frac{1}{(n+\square+1)^p} \sim \frac{p}{(n+\square)^{p+1}}$  as  $n \rightarrow \infty$  so  $\delta_n \sim \frac{1/(n+k)^{p+1}}{1/(n+1+k)^{p+1}} - 1 = \left(1 + \frac{1}{n+k}\right)^{p+1} - 1$ . Then, since  $(1+t)^{p+1} \sim 1 + (p+1)t$  as  $t \rightarrow 0$  we conclude  $\delta_n \sim 1 + \frac{p+1}{n+k} - 1$  as claimed.

Nicely enough, we can immediately extract  $p$  from  $\frac{1}{p+1} \approx \frac{1}{\delta_{n+1}} - \frac{1}{\delta_n} \sim \frac{n+1+k}{p+1} - \frac{n+k}{p+1}$ : but it should be noted that this requires 4 terms of the sequence to compute since  $\delta_{n+1}$  receives input from  $x_{n+3}$ . The reader is welcomed to consider whether comparing a 4-term formula to the original 3-term Shank's formula is *fair*, given that our error model  $x_n = x + \frac{C}{(n+k)^p}$  contains 4 unknown variables. In any case, when we write  $p$  hereafter, we mean the following estimator evaluated at each  $n$ ;

$$p = \frac{\delta_{n+1}\delta_n - \delta_n + \delta_{n+1}}{\delta_n - \delta_{n+1}}$$

To solve for  $x$  (equivalently, the error  $\epsilon_n$ ), I could see two approaches: either by using a clean rational function like;

$$\epsilon_n \approx \frac{p+1}{p} \frac{x_n - x_{n+1}}{\delta_n} \sim \frac{p+1}{p} \frac{Cp}{(n+k)^{p+1}} \frac{n+k}{p+1} = \frac{C}{(n+k)^p}$$

Or by using a more indirect numerator-denominator cancellation strategy;

$$\epsilon_n \approx \frac{x_n - x_{n+1}}{1 - \left(\frac{\delta_{n+1}}{\delta_n}\right)^p} \sim C \frac{1/(n+k)^p - 1/(n+k+1)^p}{1 - \left(\frac{n+k}{n+k+1}\right)^p} = \frac{C}{(n+k)^p}$$

These two approaches give rise to, respectively, two  $p$ -Shanks formulas;

$$\Pi_n = x_{n+1} - \frac{1}{2} \frac{p+1}{p} \frac{x_{n+1} - x_{n+2}}{\delta_{n+1}} - \frac{1}{2} \left(1 + \frac{\delta_{n+1}}{p+1}\right)^p \frac{p+1}{p} \frac{x_n - x_{n+1}}{\delta_n}$$

$$\Psi_n = x_n - \frac{x_n - x_{n+1}}{1 - \left(\frac{\delta_{n+1}}{\delta_n}\right)^p}$$

Curiously, both end up putting  $p$ , which is a rational function of the  $\delta$ 's in the *exponent* of  $\delta$ .  $\Psi_n$  is a trivial rearrangement of  $x = x_n - \epsilon_n$  but  $\Pi_n$  requires some explanation: instead of computing  $x = x_{n+1} - \epsilon_{n+1}$  directly, we pursue further accuracy by *averaging* the two error estimates  $\epsilon_{n+1}$  and  $\left(1 + \frac{\delta_{n+1}}{p+1}\right)^p \epsilon_n$ . This is because  $\epsilon_{n+1} = \frac{C}{(n+k+1)^p} = \left(\frac{n+k}{n+k+1}\right)^p \frac{C}{(n+k)^p} = \left(1 + \frac{1}{n+k+1}\right)^p \epsilon_n$  so upon substituting in  $\delta_{n+1} = \frac{p+1}{n+k+1}$ , we arrive at two compatible estimates we can average together.

I would also like to highlight the following beautiful  $p$ -Shanks formula (where  $p$  is either known or computed dynamically from  $\delta_n$  and  $\delta_{n+1}$ ):

$$S_n(p) = x_n - \frac{p+1}{p} \frac{(x_{n+1} - x_n)^2}{x_{n+2} - 2x_{n+1} - x_n}$$

Which not only bears a striking resemblance to the rational formula for  $S_n$ , but also equals  $S_n$  as  $p \rightarrow \infty$ ! (perhaps there is some intuition here about “ $\frac{1}{n^\infty}$ ” being “geometric” in  $n$ ?) We can establish the above formula by applying the asymptotic  $\frac{1}{(n+\square)^p} - \frac{1}{(n+\square+1)^p} \sim \frac{p}{(n+\square)^{p+1}}$  two times to the denominator and once to the numerator to see;

$$\frac{(x_{n+1} - x_n)^2}{x_{n+2} - 2x_{n+1} - x_n} \sim \frac{C^2 p^2 / (n+k)^{2p+2}}{C p (p+1) / (n+k)^{p+2}} = \frac{p}{p+1} \frac{C}{(n+k)^p} \approx \frac{p}{p+1} \epsilon_n$$

As claimed: just solve for  $x$  in terms of  $x_n$  and  $\epsilon_n$ .

*Exercise:* Let's talk about the denominator of  $S_n$ . If  $x_{n+2} - 2x_{n+1} - x_n = 0$  holds identically for all  $n$ , what can you conclude about the sequence  $x_n$ ? (Look at ratios or look at the characteristic polynomial).

Shanks is called a  $\Delta$ -squared process: explain why by expanding  $\frac{(\Delta x_n)^2}{\Delta^2 x_n}$  where  $\Delta x_n = x_{n+1} - x_n$  is the forward difference operator. What can you conclude about  $x_n$  when  $\delta_n = 0$  vanishes identically? How would you write  $\delta_n$  in  $\Delta$  notation?

Finally, it's not hard to see that by flipping a few  $-$ 's in the formulas provided above, one can adapt the  $p$ -Shanks transform to *alternating* series where  $\epsilon_n = C \frac{(-1)^n}{(n+k)^p}$ . However, unlike in the geometric model where  $\epsilon_n \sim C\alpha^n$  and  $\epsilon_n \sim C(-1)^n \alpha^n$  are indistinguishable, the modifications to  $\Pi_n, \Psi_n, S_n(p)$  *fundamentally* change their convergence properties.

For example,  $\delta_n$  would update to;

$$\delta_n^{\text{Alt}} = -\frac{x_{n+1} - x_n}{x_{n+2} - x_{n+1}} - 1 = -\frac{-(-1)^n C / (n+k+1)^p - (-1)^n C / (n+k)^p}{(-1)^n C / (n+k+2)^p - (-1)^n C / (n+k+1)^p} - 1 \sim \frac{p}{n+k}$$

Where instead of a  $0-0$ -type limit in the numerator, we end up with something like  $\sim \frac{2}{(n+k)^p}$  divided by  $\sim \frac{2}{(n+k+1)^p}$  which has exponent  $p$  instead of  $p+1$ . From here, the remaining modifications take the form;

$$\Psi_n^{\text{Alt}} = x_n + \frac{x_{n+1} - x_n}{1 + \left(\frac{\delta_{n+1}^{\text{Alt}}}{\delta_n^{\text{Alt}}}\right)^p} \sim x$$

$$\Pi_n^{\text{Alt}}(n) = x_{n+1} - \frac{3}{4} \frac{x_{n+1} - x_{n+2}}{2} + \frac{1}{2} \left(1 + \frac{\delta_{n+1}}{p+0}\right)^p \frac{x_n - x_{n+1}}{2}$$

Which looks quite different from what we started with!

## Numerical Experiments

To compare the performance of the series acceleration methods discussed so far, here are the results of each applied to two of the “typical use-cases” mentioned in the Introduction:

| $n$ | $x_n$            | $S_n$            | $S_n(p)$         | $\Pi_n$          | $\Psi_n$         | Sequence 1: $x_n = \frac{6}{\pi} \sum_{k=1}^n \frac{1}{k^2}$ |
|-----|------------------|------------------|------------------|------------------|------------------|--------------------------------------------------------------|
| 1   | 1.9098593        | 2.769296         | 3.5809862        | <b>3.1739729</b> | <b>3.1189265</b> |                                                              |
| 5   | 2.7952913        | 2.9952552        | <b>3.1931787</b> | <b>3.1436834</b> | <b>3.1404127</b> |                                                              |
| 10  | 2.9598383        | <b>3.0586597</b> | <b>3.1571379</b> | <b>3.1419864</b> | <b>3.1413821</b> | Converges to: $\frac{6}{\pi} \frac{\pi^2}{6} = \pi =$        |
| 20  | <b>3.0484472</b> | <b>3.0971933</b> | <b>3.145889</b>  | <b>3.1416546</b> | <b>3.1415606</b> | <b>3.14159265...</b>                                         |
| 25  | <b>3.0667058</b> | <b>3.1055661</b> | <b>3.1443998</b> | <b>3.1416259</b> | <b>3.1415755</b> |                                                              |
| 50  | <b>3.1037749</b> | <b>3.1230515</b> | <b>3.1423245</b> | <b>3.1415972</b> | <b>3.1415903</b> |                                                              |
| 100 | <b>3.1225892</b> | <b>3.1321846</b> | <b>3.1417796</b> | <b>3.1415933</b> | <b>3.1415923</b> |                                                              |

| $n$ | $x_n$           | $S_n$             | $S_n(p)$          | $\Pi_n$           | $\Psi_n$          | Sequence 2: $x_n = \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n}\right)^3$ |
|-----|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------------------------------------------------------------|
| 1   | 1               | 0.40081522        | -0.12686464       | <b>0.21778149</b> | <b>0.27711472</b> |                                                                         |
| 5   | 0.36            | <b>0.2949017</b>  | <b>0.2313823</b>  | <b>0.2493644</b>  | <b>0.25084968</b> |                                                                         |
| 10  | 0.3025          | <b>0.27371869</b> | <b>0.24519568</b> | <b>0.24993347</b> | <b>0.25015194</b> | Converges to: $\int_0^1 t^3 dt = \frac{1}{4} =$                         |
| 20  | <b>0.275625</b> | <b>0.2621819</b>  | <b>0.24877703</b> | <b>0.2499944</b>  | <b>0.25002381</b> | <b>0.25000000...</b>                                                    |
| 25  | <b>0.2704</b>   | <b>0.25979693</b> | <b>0.24921419</b> | <b>0.24999756</b> | <b>0.25001282</b> |                                                                         |
| 50  | <b>0.2601</b>   | <b>0.25494956</b> | <b>0.24980187</b> | <b>0.24999983</b> | <b>0.25000178</b> |                                                                         |
| 100 | <b>0.255025</b> | <b>0.25248744</b> | <b>0.24995024</b> | <b>0.24999999</b> | <b>0.25000024</b> |                                                                         |

And so it is abundantly clear that using the  $p$ -series error model (ie  $S_n(p), \Pi_n, \Psi_n$ ) significantly improves convergence when the error in  $x_n$  is, unsurprisingly, of  $p$ -series type. Interestingly, this advantage immediately *vanishes* when we turn to the alternating case:

| $n$ | $x_n$             | $S_n$             | $\Pi_n^{\text{Alt}}$ | $\Psi_n^{\text{Alt}}$ | Sequence 3: $x_n = \sum_{k=1}^n \frac{(-1)^{(k+1)}}{k}$ |
|-----|-------------------|-------------------|----------------------|-----------------------|---------------------------------------------------------|
| 1   | 1                 | <b>0.7</b>        | <b>0.70833333</b>    | <b>0.7</b>            |                                                         |
| 5   | <b>0.78333333</b> | <b>0.69358974</b> | <b>0.69404762</b>    | <b>0.69358974</b>     |                                                         |
| 10  | <b>0.64563492</b> | <b>0.69306575</b> | <b>0.69298341</b>    | <b>0.69306575</b>     | Converges to: $\lim_{x \rightarrow 1^-} \ln(1+x) =$     |
| 20  | <b>0.6687714</b>  | <b>0.69313464</b> | <b>0.69312205</b>    | <b>0.69313464</b>     | $\ln(2) = \mathbf{0.69314718...}$                       |
| 25  | <b>0.7127475</b>  | <b>0.69315389</b> | <b>0.6931606</b>     | <b>0.69315389</b>     |                                                         |
| 50  | <b>0.68324716</b> | <b>0.69314627</b> | <b>0.69314535</b>    | <b>0.69314627</b>     |                                                         |
| 100 | <b>0.68817218</b> | <b>0.69314706</b> | <b>0.69314694</b>    | <b>0.69314706</b>     |                                                         |

Indeed, not only does  $\Pi_n^{\text{Alt}}$  get comparatively worse than  $\Psi_n^{\text{Alt}}$ , but both do not demonstrate any advantage over ordinary shanks  $S_n$ . Curiously, it even seems like  $S_n$  and  $\Psi_n^{\text{Alt}}$  share exactly the same digits!

I do not have an explanation for why the alternating case fails *so* much worse for the  $p$ -Shanks method, but as for the second observation about  $S_n = \Psi_n^{\text{Alt}}$ , it's not hard to check *algebraically* that this holds true when the sequence  $x_n$  is of the form;

$$x_n = c + d \sum_{k=1}^n \frac{(-1)^{k+1}}{ax+b} \implies p = \left( \frac{1}{\delta_{n+1}^{\text{Alt}}} - \frac{1}{\delta_n^{\text{Alt}}} \right)^{-1} = 1 \quad \forall n$$

For example,  $\ln(2)$  is of this form as is  $\frac{\pi}{4} = \sum_{n \geq 0} \frac{(-1)^n}{2n+1}$ . Obviously the two formulas are not equal in general, but it certainly is not obvious why this particular structure for  $x_n$  is what gives the coincidence!

It's an interesting question whether similar Shanks generalizations can be furnished from other error models. For example, recall that the convergence of an  $n$ -th order Taylor polynomial to a smooth function  $f$  has error (loosely speaking) governed by the Lagrange bound  $\epsilon_n \sim \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^n$  which is *neither* a geometric series in  $n$  nor a  $p$ -series in  $n$ . Or for instance, certain number-theoretic sums of primes or sums over twin-primes converge at a rate of  $O(\ln n)$  or  $O(\ln \ln n)$ . In the case of Taylor, the error converges rapidly anyway, and in the case of the distribution of primes, the log error model is dubiously accurate for most reasonably small  $n$ , so I have doubts as to whether such an approach would be helpful. But, the reader is encouraged to explore and experiment!